12-16 第1篇

阅读理解（说明文） Why Do Onions Make Us Cry?

① Anyone who has ever stood next to a kitchen chopping-board with a small sharp knife and half a fresh onion knows the quiet, slightly funny moment very well. At first, nothing. Your eyes are fine. You start slicing. For **eight or nine seconds**, the onion simply gives off a familiar sweet-sharp smell. Then, suddenly, without anyone telling them to, your eyes fill with warm clear tears. You cannot stop them. You are not sad. The onion is not spicy on your tongue. Yet, in under ten seconds, a small ordinary vegetable has quietly forced you into a small ordinary cry. This tiny kitchen mystery has a careful chemistry answer — and the answer is actually one of the most beautiful small self-defence systems in the whole plant world.

② **Finding one: the onion does not always have the "crying chemical" inside it.** When an onion is sitting quietly in the fridge, it contains three main things relevant to our story: a sulfur-rich building block called **isoalliin**, an enzyme (a kind of biological scissors) called **alliinase**, and a very special **second enzyme** called **lachrymatory-factor synthase (LFS)**. In an intact onion, these three things are **kept apart in different tiny compartments inside the cells**. The onion is, in this state, completely safe to touch, to smell, and to hold in your hand. **Nothing happens until the cells are broken.**

③ **Finding two: the tears only start when you cut.** The moment your knife slices through the onion, you tear open thousands of tiny plant cells, and the three things — isoalliin, alliinase, and LFS — are suddenly thrown together into one small wet reaction area. Alliinase first cuts isoalliin into a smaller, angry sulfur compound. LFS, the second enzyme, then quickly rearranges it into a sharp, very light gas called **syn-propanethial-S-oxide** — the famous "onion tear gas." This gas rises from the cut surface, **drifts upward**, and reaches your eyes in roughly **2—10 seconds**. (Japanese scientists at the **House Foods laboratory, 2002**, were the first to isolate LFS and prove this was the real "tear factor" — they won an Ig Nobel Prize in **2013** for their work.)

④ **Finding three: your eye does not cry because it is hurt.** Your eye cries because it is **cleaning**. The thin sulfur gas reaches the front of the eye and dissolves in the thin layer of tears on the cornea. A small reaction forms a tiny amount of **weak sulfuric acid** — extremely small, far too weak to hurt the eye. But the **trigeminal nerve**, the large nerve that watches the front of your face for any chemical stress, is very, very sensitive, and it sends an instant signal: *"Something mildly irritating has arrived — flush it out."* The **lacrimal glands**, just above your eyes, immediately release extra tears. The tears wash the sulfur compound away. Within **30 to 60 seconds** of you stopping the cutting, the gas disperses and the tears stop. There is no injury. Your eye simply did its cleaning job.

⑤ **Why does the onion bother to do this?** The answer is quietly clever: the onion is a bulb — a seed-like underground package of food — and in the wild, hungry animals (rabbits, rats, even insects) love to dig it up and eat it. A fresh onion cannot run away. But it can, the moment it is bitten or cut, release a sharp stinging gas into the attacker's eyes, nose, and mouth. Most animals, after one attempt, leave it alone. **The onion's tear gas is not there to annoy your mother in the kitchen; it is an ancient, small, non-lethal alarm system that has kept onions alive for 6,000 years of farming and many more years before.**

⑥ **How to cut an onion without crying (three science-based tricks).** Knowing the chemistry, we can quietly cheat it:

Trick	Why it works	How well
Chill the onion for 20 minutes in the fridge before cutting	Cold slows enzyme reactions; alliinase and LFS work more slowly at low temperature, producing less tear gas	Very helpful
Cut the onion under a thin stream of cold water, or keep the chopping board wet	Water dissolves the sulfur gas before it rises to your eyes	Helpful
Use a very sharp knife; do NOT crush the onion	A clean cut breaks fewer cells than a dull knife that crushes; fewer broken cells = less reaction	Very helpful
Breathe through your mouth, keep your nose away from the cut	Reduces the amount of gas reaching the eye-area through the air currents created by your breathing	Mild help
Chewing gum while cutting	Breathing pattern changes and saliva increases; some small home studies suggest mild reduction — no strong science yet	Possibly mild

⑦ **What about "tearless onions"?** In **2008**, agricultural scientists in **New Zealand and Japan** used careful traditional plant breeding (not genetic modification) to produce onions with **very low LFS activity**. These "tearless onions" still have isoalliin and alliinase, so they still taste like onions — but they produce almost no syn-propanethial-S-oxide gas. They appeared in some supermarkets around 2014. Many chefs, interestingly, **prefer the crying kind**: they say the small sulfur compounds, which break down further when cooked, give the deeper sweet flavour of a well-caramelised onion. A tearless onion, in other words, is a slightly quieter onion in every sense.

⑧ **Is onion tearing bad for you?** No — and sometimes, in a small way, good. A **2021 Tel Aviv University study** of **74 adults** compared tear samples after onion cutting, after sad feelings, and from relaxed rest. The onion tears were chemically **different** from emotional tears: they contained **more antibacterial enzymes** and a **simpler salt profile**. Onion tears, it turns out, give the eye surface a small free wash — which is why some eye doctors quietly note that a kitchen full of onion work rarely harms a healthy eye.

(Contact-lens wearers, of course, are a small exception: the tears may briefly fog lenses.)

⑨ **The quiet lesson.** Next time you cut an onion in a kitchen and the warm clear tears come — for no reason you can name — remember: you are not sad. You are not hurt. You are simply standing inside a **6,000-year-old quiet chemistry experiment** that a small underground bulb is performing, one last time, in its own small self-defence. Your eye is doing a small clean. Your knife is doing a small cut. And the onion, which cannot speak and cannot run, is using the only kind of voice it has — a small sulfur whisper that, for two or three seconds in a winter kitchen, turns water into a soft silver line on your cheek.

注释： isoalliin 异蒜氨酸；alliinase 蒜氨酸酶；LFS 催泪因子合成酶；syn-propanethial-S-oxide 丙硫醛-S-氧化物；lachrymatory 催泪的；trigeminal 三叉神经的；lacrimal 泪腺的；caramelise（使）焦糖化

根据短文内容，选择最佳答案。

1. According to Paragraph ②, why does a whole uncut onion not make us cry?

A. Because the whole onion has no sulfur.
B. Because onions are always dry and harmless.
C. Because a whole onion contains isoalliin, the enzyme alliinase, and a second enzyme called lachrymatory-factor synthase (LFS), but these three things are kept apart in different tiny compartments inside the cells; in this state the onion is completely safe to touch, smell and hold, and nothing happens until the cells are broken.
D. Because whole onions emit an anti-tear chemical.

2. What happens step-by-step between cutting the onion and your eyes starting to cry, according to Paragraphs ③ and ④?

A. Cutting releases a magical smell that hypnotises the eye.
B. The knife tears thousands of tiny plant cells so that isoalliin, alliinase and LFS are thrown together in one wet reaction area: alliinase cuts isoalliin into a smaller angry sulfur compound, LFS rearranges it into a sharp light gas called syn-propanethial-S-oxide that drifts upward and reaches the eyes in 2—10 seconds; this gas dissolves in the thin tears on the cornea forming a very weak and harmless amount of sulfuric acid; the highly sensitive trigeminal nerve senses this mild chemical stress and signals the lacrimal glands to release extra tears, and the tears wash the sulfur compound away — within 30—60 seconds of stopping the cutting, the gas disperses and the tears stop.
C. Onion dust enters the eye and causes deep injury.
D. The eye cries because it is emotionally moved by the onion.

3. What does the writer identify as the evolutionary purpose of the onion's tear gas, and what is its nature?

A. The tear gas exists to annoy human cooks and make them cry.
B. The onion is a bulb — a seed-like underground package of food — that cannot run away; the tear gas, released only when the bulb is bitten or cut, is an ancient, small, non-lethal alarm system that stings the attacker's eyes, nose and mouth so that most animals leave it alone after one attempt, keeping onions alive for 6,000 years of farming and many more years before.
C. The tear gas is meant to kill insects completely.
D. The tear gas is a side-effect with no purpose.

4. Which of the following statements about the three "science-based tricks" and "tearless onions" is TRUE according to Paragraphs ⑥ and ⑦?

A. Chilling the onion for 20 minutes before cutting makes the tear gas worse because cold sharpens smell.
B. A dull knife is better because it breaks fewer cells than a sharp one.
C. Chewing gum is proven to completely stop onion tears.
D. Chilling the onion for 20 minutes in the fridge before cutting slows enzyme reactions because alliinase and LFS work more slowly at low temperature and produce less tear gas; keeping the chopping board wet dissolves the sulfur gas before it rises; using a very sharp knife breaks fewer cells than a dull knife that crushes; and "tearless onions" were produced in 2008 by New Zealand and Japanese scientists using traditional plant breeding (not GM), with very low LFS activity, so they taste like onions but release almost no gas — many chefs still prefer the crying kind because the small sulfur compounds give deeper sweet flavour when caramelised.

5. What is the main idea of the final paragraph?

A. Cooks should avoid onions forever because the tears are shameful.
B. Onion tears are physically harmful to all eyes and should be stopped at all costs.
C. Onions are worse than any other vegetable in the kitchen.
D. When the warm clear tears come during onion cutting, the cook is not sad and not hurt but standing inside a 6,000-year-old chemistry experiment performed, one last time, by a small underground bulb in self-defence; the eye is doing a small clean, the knife a small cut, and the onion — unable to speak or run — uses the only kind of voice it has, a small sulfur whisper that briefly turns water into a soft silver line on the cook's cheek.

答案与解析

1.
【答案】C

【题型】细节理解题
【定位】第②段完整洋葱
【解析】"三物质分隔 / 细胞不破则无反应"直引。A、B、D 均错。故选 C。

2.
【答案】B
【题型】细节理解题
【定位】第③④段反应链与眼部清洗
【解析】"细胞破裂 → 酶切 → LFS → 气体上升 → 弱酸 → 三叉神经 → 泪腺洗"直引。A、C、D 均错。故选 B。

3.
【答案】B
【题型】推理判断题 / 细节理解题
【定位】第⑤段进化目的
【解析】"鳞茎 / 被咬触发 / 非致命报警 / 6000 年农耕"直引。A、C、D 均错。故选 B。

4.
【答案】D
【题型】细节理解题
【定位】第⑥⑦段科学小技巧 + 无泪洋葱
【解析】"冷藏慢酶 / 湿板 / 锋利刀 / 2008 无泪品种 / 风味更浅"直引。A、B、C 均错。故选 D。

5.
【答案】D
【题型】主旨大意题
【定位】第⑨段结尾
【解析】"6000 年化学实验 / 洋葱唯一声音 / 银线流过脸颊"主旨。A、B、C 均偏离。故选 D。